

ICU Nursing Procedures Manual

Standard Operating Procedures for Critical Care Nursing

MediAssist Health Network

Nursing Education & Critical Care

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Access: Nurses, Doctors, Admin



This manual sets out standard operating procedures (SOPs) for common ICU nursing tasks at MediAssist hospitals. Each SOP lists pre-requisites, numbered steps, monitoring parameters and documentation requirements. Procedures must be performed only by staff assessed as competent for the task.

SOP 1 — Central Venous Catheter (CVC) Care

Frequency

Dressing change **every 72 hours**, or sooner if the dressing is soiled, loose or no longer occlusive.

Equipment checklist

- Sterile gloves and sterile dressing pack
- Chlorhexidine 2% in 70% alcohol solution
- Sterile drape
- Transparent semi-permeable dressing
- Date label

Procedure

- 1** Perform hand hygiene and don personal protective equipment (PPE).
- 2** Remove the old dressing and discard in the yellow (biomedical) bin.
- 3** Inspect the insertion site for redness, swelling, or discharge.
- 4** Perform hand hygiene again and apply sterile gloves.
- 5** Clean the site with chlorhexidine using a concentric (inside-out) motion.
- 6** Allow to air-dry for at least 30 seconds — do not fan or blow.
- 7** Apply the new transparent dressing.
- 8** Label the dressing with the date and your initials.
- 9** Document the procedure and site assessment in the care record.

Escalate

Notify the doctor immediately for: purulent discharge, fever > 38.5 °C with no other source, or tracking erythema extending > 2 cm from the insertion site.

SOP 2 — Mechanical Ventilator Management

Initial settings

Parameter	Starting Value
Mode	Assist-control / Volume-control (AC/VC)
Tidal volume	6–8 mL/kg ideal body weight
Respiratory rate	12–16 breaths/min
FiO ₂	Start 1.0, then titrate to SpO ₂ 94–98%
PEEP	5 cmH ₂ O (default)

Hourly monitoring

- SpO₂, end-tidal CO₂ (EtCO₂)
- Peak and plateau airway pressures
- Delivered tidal volume
- Ventilator alarm status

Alarm response

Alarm	Immediate Action
High pressure (> 40 cmH ₂ O)	Check for secretions / kinking / coughing → suction if indicated → call doctor if it persists
Low SpO ₂ (< 90%)	Increase FiO ₂ → assess airway and circuit → call doctor immediately

Ventilator bundle (VAP prevention)

- Head of bed (HOB) elevation 30–45°
- Oral care with chlorhexidine every 4 hours
- Subglottic suctioning where available
- Daily sedation vacation and readiness-to-wean assessment

SOP 3 — Nasogastric Tube (NGT) Insertion & Management

Sizing

Adult: **14–16 Fr.** Paediatric: select by age-appropriate formula.

Insertion procedure

- 1 Measure from the tip of the nose to the earlobe to the xiphoid process; mark the tube.
- 2 Lubricate the tip with water-soluble gel.
- 3 Insert through the nostril, advancing gently.
- 4 Ask the patient to swallow (sips of water if permitted) as you advance.
- 5 Advance to the measured length.
- 6 Confirm position by aspirate pH < 5.5 AND chest X-ray before first feed.

Feeding & gastric residual monitoring

Situation	Action
Initiation	Start at 20 mL/hr
If GRV < 200 mL	Increase by 20 mL/hr every 4 hours
If GRV 200–400 mL	Hold feed 1 hour and recheck
If GRV > 400 mL	Hold feed; notify dietitian and doctor

Safety

GRV = gastric residual volume, checked every 4 hours. Never confirm NGT position by auscultation alone before feeding.

SOP 4 — IV Cannula Insertion

Site selection (order of preference)

Forearm → antecubital fossa → dorsum of hand. Avoid the legs except in emergency.

Cannula sizing by indication

Indication	Recommended Gauge
Blood transfusion	≥ 18G
Rapid fluid resuscitation	≥ 16G
Routine medication	20–22G
Paediatric < 5 kg	24G
Paediatric 5–20 kg	22G
Paediatric > 20 kg	20G

Failed attempt protocol

- Maximum **2 attempts per nurse**.
- After 2 failed attempts, escalate to a senior nurse or doctor.

Cannula change

Replace every **72–96 hours**, or immediately if there are signs of phlebitis (Visual Infusion Phlebitis [VIP] score ≥ 2).

SOP 5 — Patient Restraint Protocol

Principle

Restraint is a last resort to prevent imminent harm (self-extubation, falls) and must **never** be used as routine care or for staff convenience.

Authorisation

- Requires a verbal doctor's order, with a **written order within 1 hour**.
- Re-authorisation is required **every 24 hours**.

Monitoring while restrained

- Neurovascular check **every 2 hours** — circulation, sensation and movement in the restrained limb.
- Offer range-of-motion exercises every 2 hours.
- Document every check.

Documentation requirements

- Reason for restraint and alternatives tried
- Time applied and type of restraint
- All monitoring observations
- Communication with the family

SOP 6 — Endotracheal Suctioning

Suctioning clears secretions to maintain airway patency in the intubated patient. It is performed only when clinically indicated, not on a fixed routine.

Indications

- Visible or audible secretions / coarse breath sounds
- Rising peak airway pressure or falling tidal volume
- Desaturation or a sawtooth pattern on the flow-volume loop

Procedure

- 1 Hand hygiene; don gloves, apron and eye protection.
- 2 Pre-oxygenate with FiO₂ 1.0 for 30–60 seconds.
- 3 Use a closed in-line suction catheter where available; select catheter ≤ half the ETT internal diameter.
- 4 Insert without suction; apply suction on withdrawal for no more than 10–15 seconds.
- 5 Limit to two passes; allow recovery between passes.
- 6 Return ventilator settings to baseline and reassess.
- 7 Document secretion colour, volume and the patient's tolerance.

Safety

Stop suctioning immediately for SpO₂ < 90%, bradycardia, or a new arrhythmia, and re-oxygenate the patient.

SOP 7 — Pressure Injury Prevention

Risk assessment

Assess every ICU patient using the **Braden Scale** on admission and at least every 24 hours. A score ≤ 18 indicates increased risk.

Preventive bundle

- **Repositioning** — turn the patient every 2 hours and document the position.
- **Support surface** — use a pressure-redistributing mattress for high-risk patients.
- **Skin inspection** — inspect bony prominences each shift.
- **Moisture management** — keep the skin clean and dry; apply barrier cream as needed.
- **Nutrition** — refer at-risk patients to the dietitian.

Staging quick reference

- **Stage 1** — non-blanching erythema with intact skin.
- **Stage 2** — partial-thickness skin loss; a shallow open ulcer.
- **Stage 3** — full-thickness skin loss with fat visible.
- **Stage 4** — full-thickness tissue loss with muscle or bone visible.

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